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Completed
13/10/26

1. Network Commands (Linux & Windows)

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.22631.4460]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ladch>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:
   IPv4 Address. . . . . . . . . . . : 192.168.1.10
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1

C:\Users\ladch>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=24ms TTL=115
Reply from 8.8.8.8: bytes=32 time=24ms TTL=115
Reply from 8.8.8.8: bytes=32 time=24ms TTL=115
Reply from 8.8.8.8: bytes=32 time=24ms TTL=115
```

27/7

20/07

Experiment - 1

Basic Networking Commands:

Aim:

To study the various network commands used in linux and windows.

Command:

1. arp-a :- address resolution protocol

⇒ It is short form of ~~all TCP/IP commands~~. It will show the IP address of computer along with IP address and MAC address of your router.

O/P:

-gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3

? (172.16.8.1) at <incomplete> on enp0 s3

2. hostname :

⇒ It a simplest of all TCP/IP commands. It simply displays name of your computer.

O/P:

fedora

3. ipconfig /all :

⇒ It displays detailed configuration information about your TCP/IP connection including router, gateway, DNS, DHCP,

and type of Ethernet adapter on your system.

O/P:

```
$ip addr
1o . < LOOPBACK, UP, LOWER_UP> mtu 65536
    inet 127.0.0.1/8 scope host lo
    inet ::1/128 scope host
```

4. nbtstat -a:

⇒ It helps to solve problems with NetBIOS name resolution (Nbt stands for NetBIOS over TCP/IP)

O/P:

-a (adapter status) Lists the remote machine's name table given its name

-A (Adapter status) Lists the remote machine's name table given its IP address.

nbtstat -n

O/P: NetBIOS Local name Table

Name	Type	Status
DESKTOP-MR039M3 <20>	UNIQUE	Registered
DESKTOP-MR039M3 <00>	UNIQUE	Registered
WORKGROUP <00>	GROUP	Registered

netstat:

⇒ displays variety of statistics about a computers active TCP/IP connections.

O/P:

Proto	RefCnt	Flags	Type	State	I-Node	Path
unix	2	[ ]	DGRAM	CONNECTED	50445	
unix	3	[ ]	STREAM	CONNECTED	24535	

6. nslookup:

⇒ (name serva lookup) is a tool used to perform DNS lookups on linux. It displays DNS details, such as IP addres of particular computer, the MX records for a domain or NS servers of a domain.

O/P:

```
* nslookup www.google.com
    Serva :  127.0.0.53
    Address :  127.0.0.53 #53
Non-authoritative answer:
Name :  www.google.com
Address :  142.251.156.119
```

7. pathping:

⇒ It is unique to Window's, and is basically a combination of ping and Tracert commands.

O/P:

Usage : pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
[-p period] [-q num_queries] [-w timeout]
[-4] [-6] target_name

8. ping:

⇒ (Packet INternet Groper) command is best way to test connectivity b/w 2 nodes. It uses ICMP (Internet Control Message Protocol) to communicate to other device.

O/P:

ping hostname (ping localhost)

ping statistics for 142.251.152.119:

packets: sent = 4, recieved = 4, host = 0 (0% Los

9. Route:

⇒ It is used to show/manipulate the IP routing table. It is primarily used to setup static routes to specific host or networks via an interface.

O/P:

Kernal IP routing table

Destination	Gateway	Genmask	Flags	Metic	Ref	Use	IF
default	gateway	0.0.0.0	UG	100	0	0	en
10.0.2.0	0.0.0.0	255.255.255.0	U	100	0	0	enp

Ex No: 2

Study of Different types of Network Cables

Aim:

To study different types of network cable.

a) To understand different types of network cable.

1. Unshielded twisted Pair (UTP) cable
2. Shielded Twisted Pair (STP) cable
3. Coaxial cable
4. Fibre optic cable

Cable Type	Category	Max data transmission	Advantage/ Disadvantage	Application/use
UTP	Category 3	10 bps	Advantage: Cheap in cost Easy to install as they have smaller overall diameter. Disadvantage: More prone to (EMI) Electro magnet interference and noise.	* 10 Base T-ethernet
	Category 5	upto 100 mbps		* Fast ethernet gigabit ethernet
	Category 5e	1 Gbps		* Fast ethernet gigabit ethernet.
	Category 6, 6a	10 Gbps	Advantage: * Shielded * Faster than UTP. * Less susceptible to noise and interference	* Gigabit ethernet 10G ethernet (SSN widely used in data centres.

SSTP	category 7	10Gbps	Disadvantage: *Expensive *Greater installation effort.	* Gigabit ethernet, PCI Ethernet (LAN)
Coaxial cable	RG-6 RG-5a RG-11	10-100 mbps	Advantage: *High bandwith * Immune to interfare * Low loss bandwidth *versatile Disadvantage: *Limited distance *Cost * Bulky in size	* Speed of sign is 500m. * Television network. * High speed internet connec
Fibre optics cable	Single mode, multi mode	100 Gbps.	Advantage: *High speed * High bandwidth *High security * Long distance Disadvantage: *Expensive * Required skilled installa	* Maximum distance of fibre optics cables is arou 100 metre.

Results:

Thus, the study of different type of network cable studied successfully.

Aim:

To study the cisco packet tracer simulation tool, understand its user interface design a simple network and analyze the behaviour of Hub, switch and peer-to-peer.

Cisco Packet tracer:

It is a network simulation and modeling tool. It allows us to execute and test computer networks without using real networking equipment we can add PCs, switches, hubs, routers and other devices and observe how packet travels between them.

① Peer-to-peer:

Two computers communicate directly.

A peer to peer network was created by connecting 2 PCs directly. IP addresses were assigned to both PCs in the same network. The connectivity between the PCs was tested using ping command. Successful replies were received, confirming that the two PCs could communicate directly with each other.

② Network using hub:

Connected multiple computers & forwards received data to all ports. A network was created by connecting 3 PCs to a Hub using copper straight through cables. IP addresses were assigned to all PCs. When data was sent from one PC to another the hub forwarded the packet to all connected ports. The destination PC received the packet successfully.

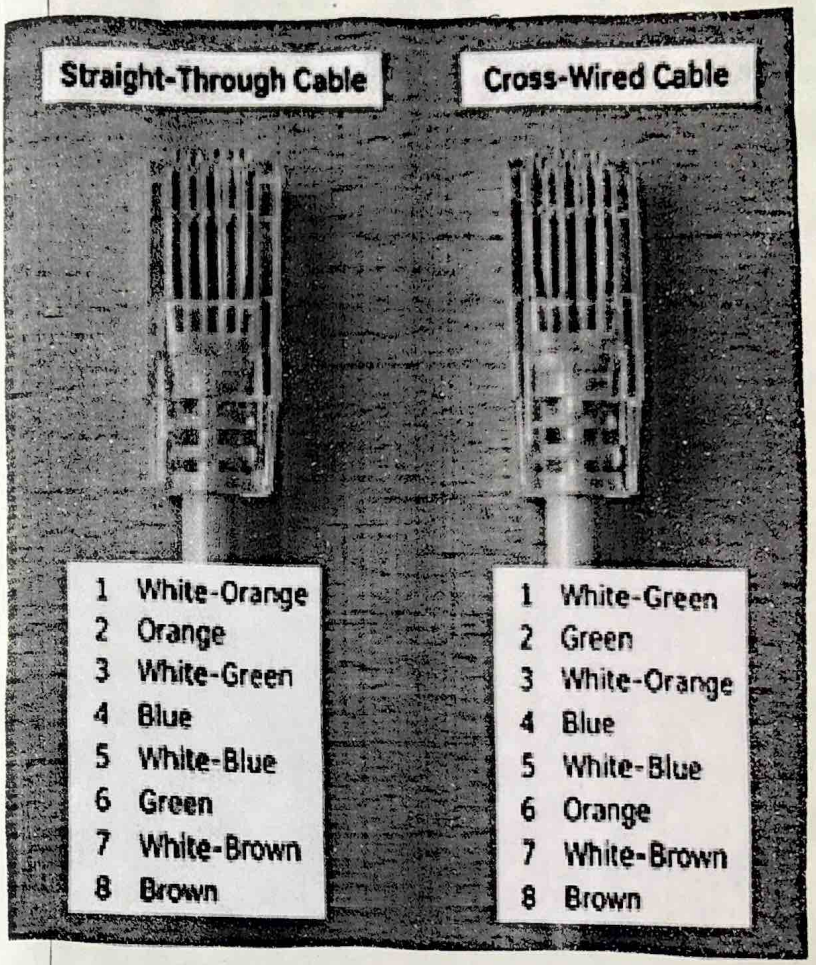
③ Networking switch:

Connects computer and forwards data to required destination.

A network was created by connecting 3 PCs to a switch using copper straight through cable. IP addresses were assigned to all PC's and connectivity was tested using ping. The switch forwarded the packet to appropriate destination.

④ Hub and Switch:

Both hub and switch were used in the network the packet forwarding behaviour of both devices was observed and compared.

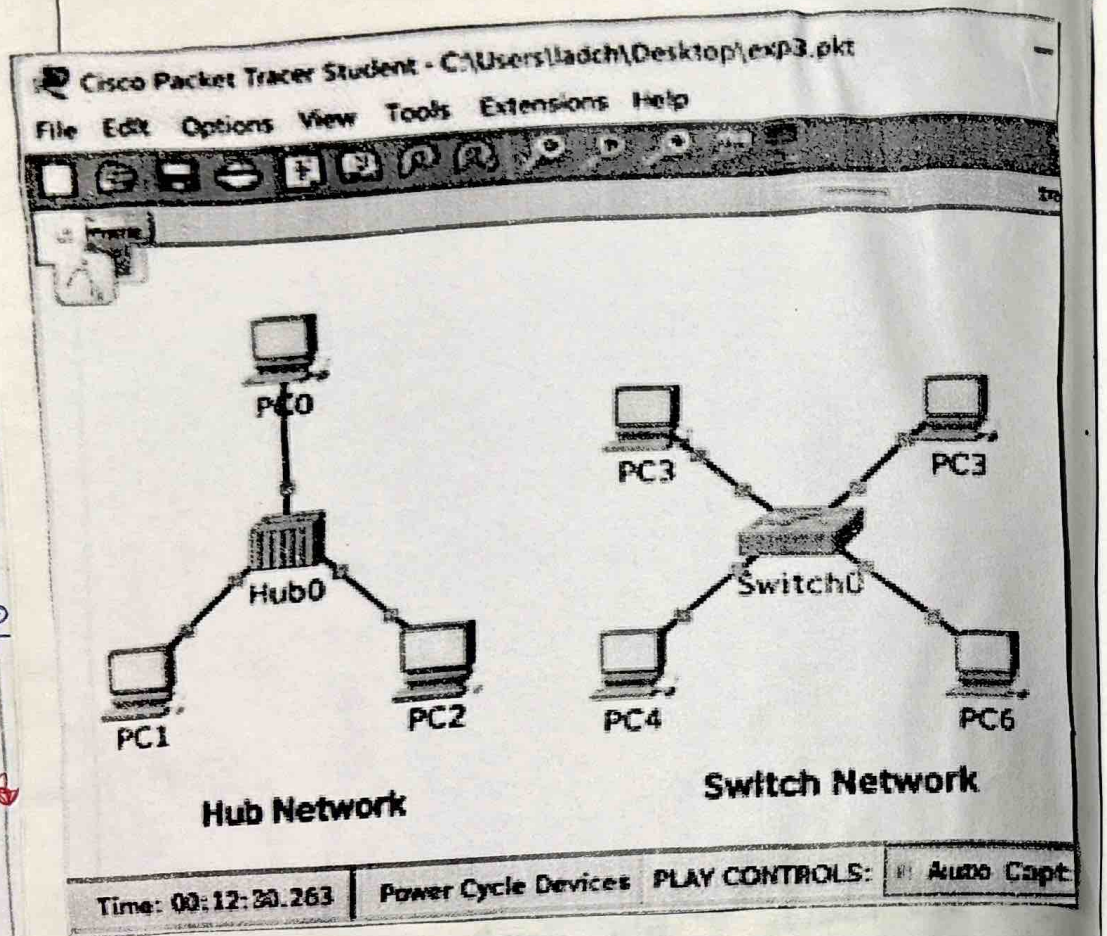


Result:

The peer to peer, hub & switch networks were successfully designed and their communication behaviour was observed using cisco packet tracer.

Ex No: 03

Study of Cisco Packet Tracer and Analysis of Network Devices.



Aim:

To study the installation and user interface of cisco packet tracer and to understand its environment for designing a simple network devices such as Hub and switch using cisco packet tracer.

Requirements:

* Cisco Packet Tracer
* 4 Generic PCs
* 1 Hub
* 1 Switch
* Copper straight-through cables

Algorithm:

1. Open Cisco Packet Tracer
2. Place 4 PCs and a Hub
3. Connect all PCs to the Hub.
4. Assign IP addresses to the PCs.
5. Send a PDU from one PC to another.
6. Observe how the hub forwards the packet.
7. Create another network using 4 PCs and a switch.
8. Assign IP addresses.
9. Send a PDU and observe the packet flow.
10. Compare Hub and Switch.

Procedure:

- Network using Hub.

1. Open packet tracer.
2. Place 4 PCs and 1 Hub.
3. Connect all PCs to the hub using copper straight-through cable.
4. Configure the IP addresses.

PC	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

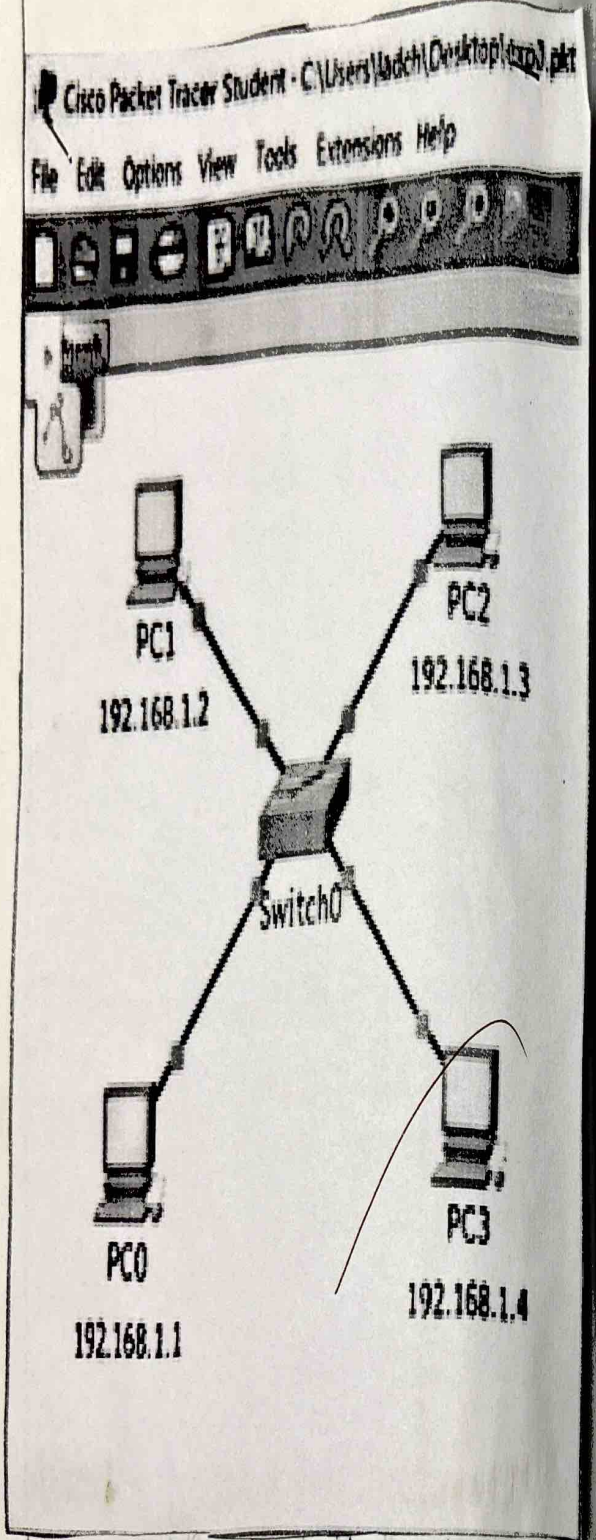
5. Select Add Simple PDU
6. Click PC0 and then PC1
7. Observe the packet flow.

Network using switch

1. Place 4 PCs and 1 switch
2. Connect all PCs to the switch
3. Align IP addresses.
4. Send a PDU from PC0 to PC1
5. Observe the packet flow.

Result:

Thus, cisco packet tracer was studied successfully. The behaviour of Hub and switch was observed and compared.



Ex No: 4

LAN Setup Using Switch:

Aim:

To setup and configure a LAN using a switch and Ethernet cable and check the connectivity between computers.

Requirements:

- 4 PCs
- 1 Switch
- 4 Ethernet Cables

Algorithm:

1. Connect 4 PCs to a switch.
2. Assign IP addresses.
3. Check the connection using ping.
4. Share a folder between computers.
5. Check whether the shared folder can be accessed.

Procedure:

1. Connect four PCs to switch using Ethernet cables.
2. Assign the IP addresses.
3. Open command prompt.
4. Check the IP using: ipconfig
5. From PC1, check the connection to PC2 using: Ping 10.1.

Result:

Thus, a LAN was successfully created and configured using a switch and ethernet cables. Connectivity between the computers was verified using the ping comm

Sample Output:

No	Protocol	Source	Destination
1	ARP	192.168.1.10	Broadcast
2	DNS	192.168.1.10	8.8.8.8
3	TCP	192.168.1.10	142.x.x.xx.xx
4	HTTP	192.168.1.10	Server

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.1	192.168.1.10	DNS	78	Standard
2	0.012345	192.168.1.10	8.8.8.8	DNS	90	Standard
3	0.023456	192.168.1.10	8.8.8.8	ICMP	98	Echo (pi
4	0.045678	8.8.8.8	192.168.1.10	ICMP	98	Echo rep
5	0.045678	192.168.1.10	192.168.1.1	TCP	54	49321→44
5	0.016899	8.8.8.8.1	8.8.56.1.1	TCP	54	49321→44
6	0.000005	192.168.1.10	8.8.8.1.1	TCS	54	49321→44
7	0.020505	192.168.1.10	192.168..3	TOS	54	49321→44
8	0.010609	192.168.1.10	192.168.1.10	TOS	54	49321→44
9	0.015608	192.8.8.1	8.8.68.1.1	TCP	54	49321→44

> Frame 3: 98 bytes on wire (784 bits), 98 bytes captured (784 bit
> Ethernet II, Src: Dell_3a:5e:b1 (a4:ae:12:3a:5e:b1),
> Internet Protocol Version 4, Src: 192.168.1.10, Dst: 8.8.8.8
> Internet Control Message Protocol

Ex No: 05

WIRESHARK

Aim:

To study the wireshark packet capture tool and analyze packets at different layers of the protocol stack.

Procedure:

1. Open Wireshark.
2. Select the local area connection / Ethernet interface
3. Start packet capture.
4. Capture approximately 100 Packets.
5. Stop the capture.
6. Apply filters for
   - → TCP / UDP
   - → ARP
   - → DNS
   - → HTTP
   - → IP / ICMP
   - → DHCP
7. Select individual packets.
8. Observe the packet list, packet details and packet bytes panes.
9. Analyze the protocol fields.
10. Save the captured packets.

Result:

Thus, wireshark was successfully used to capture, filter and analyze network packets and their protocol fields.

IDLE Shell 3.10.6

File Edit Shell Debug Options Window Help

```
Enter 4-bit data : 1011
Hamming Code : 0110011
Received Code : 0110001
Error detected at position : 3
Corrected Code : 0110011
Corrected Data : 1011
>>>
```

Error detected at position : 12

Error corrected successfully -

Decoded Data:

HELLO

Ex No: 06

HAMMING CODE:

Aim:

To implement error detection and correction using Hamming code and verify correction capability using a test data.

Procedure:

Sender:

1. Create a sender program.
2. Read a text message from input field.
3. Convert text to binary
4. Calculate the least the required redundant parity bits.
5. Generate the hamming code.
6. Store the encoded data in a file named channel.

Receiver:

7. Read the encoded data from channel.
8. Recalculate the parity bits.
9. Determine whether an error is present.
10. If error is present, find its position.
11. Remove the redundant bits.
12. Convert corrected binary data back into ASCII/text.
13. Display the recovered message.

Result:

Thus, hamming code was successfully implemented.

Sample Output:

Enter Window Size : 3
Enter Message : HELLO
Sender Buffer:
Frame 0 : H
Frame 1 : E
Frame 2 : L

Receiver:
ACK 0
ACK 1
ACK 2

Sender :
Frame 3 : L
Frame 4 : O

Receiver :
ACK 3
ACK 4

Transmission ~~Set~~ Successful.

Ex No:7

-- SLIDING WINDOW PROTOCOL --

Aim:

To implement flow control at the Data Link layer using sliding window protocol and simulate the transfer of frames between two nodes.

Procedure:

Sender:

1. Get the window size from the user.
2. Read a text message
3. Consider one character as one frame.
4. Assign a frame number to each character
5. Create frames in the format.

[Frame No, DATA]

6. Store the frame in sender_buffer.
7. Send frames according to the window size
8. Wait for acknowledgment from the receiver.
9. Read receiver_buffer.
10. Check the Ack_number.
11. If the ACK is correct, transmit the next set of frames.
12. If NACK is received, retransmit the required frames
13. Introduce an error by manually modifying a frame /ACK number and observe the behaviour

```
Enter window size : 3
Total number of frames : 7

Sending frames...
Frame 0 sent        ACK received
Frame 1 sent        ACK received
Frame 2 sent        ACK lost  ->  Retransmitting
Frame 2 sent        ACK received
Frame 3 sent        ACK received
Frame 4 sent        ACK received
Frame 5 sent        ACK received
Frame 6 sent        ACK received

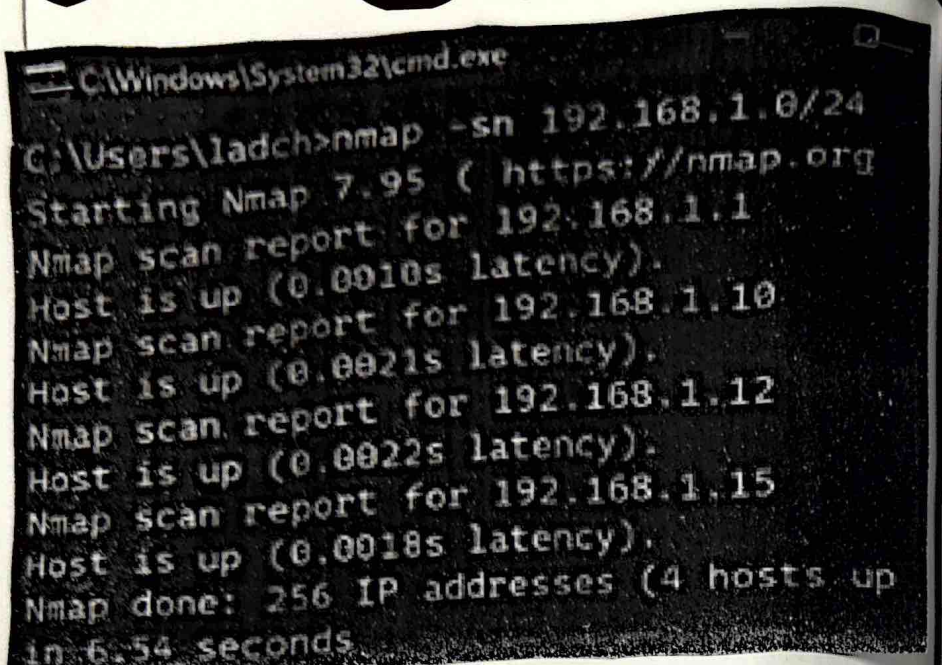
Transmission completed successfully.
```

Receiver:

14. Read frame from sender_buffer.
15. Check the frame number.
16. If correct, generate ACK.
17. If incorrect, generate NACK.
18. Store the res

Result:

Thus, sliding window protocol was successfully implemented and frame flow control using ACK/NACK was verified.



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Ex No: 08

NMAP LIVE HOST DISCOVERY

Aim:

To discover live hosts using Nmap scans based on ARP, ICMP, TCP and UDP on TryHackMe Nmap Platform.

Procedure:

1. Open Nmap room on TryHackMe.
2. Study the target network and subnet structure.
3. Identify the target IP addresses.
4. Perform ARP-based host discovery.
5. Use
   nmap -PR -sn TARGET
6. Perform ICMP Echo discovery using
   nmap -PE -sn TARGET.
7. Perform ICMP Timestamp discovery using
   nmap -PP -sn TARGET
8. Perform TCP SYN ping using
   nmap -PS23 TARGET
9. perform TCP ACK ping using the appropriate -PA option
10. Study UDP-based host discovery
11. Observe the live hosts identified by Nmap
12. Study reverse DNS lookup using -R.

Result:

Thus, Nmap was successfully used to discover live hosts using ARP, ICMP, TCP and UDP based tech

Sample Output:

PC17 ping 192.168.2.11

Reply from 192.168.211:

Bytes = 32 Time<1 ms TTL=126

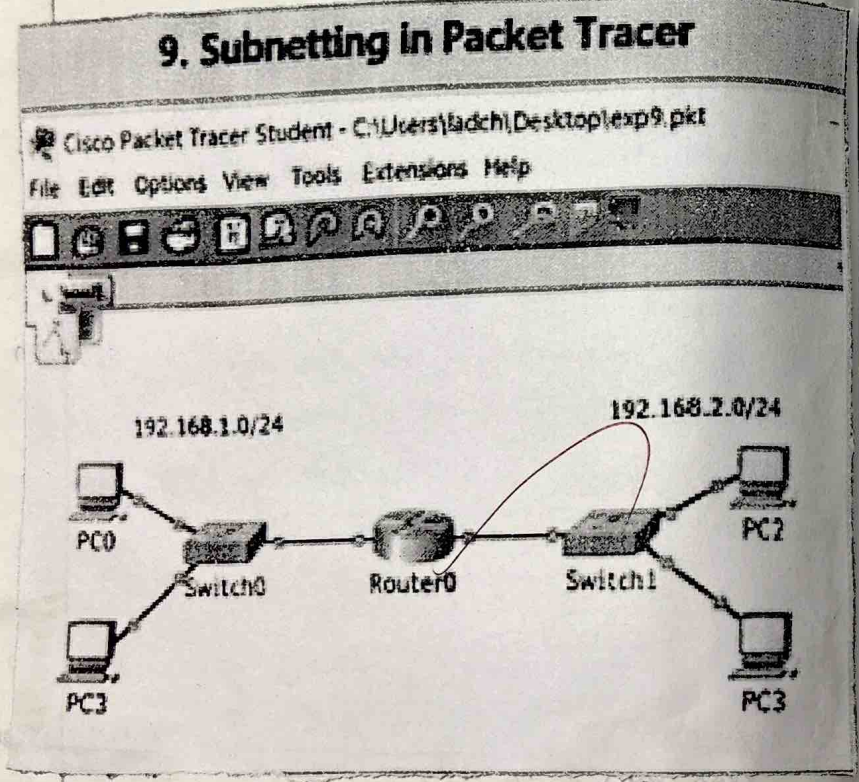
Packets: Sent = 4

Received = 4

Lost = 0

0% Packet loss

Subnet Mask: 255.255.255.224 (/27)



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Ex No: 09

SUBNETTING USING CISCO PACKET TRACER

Aim:

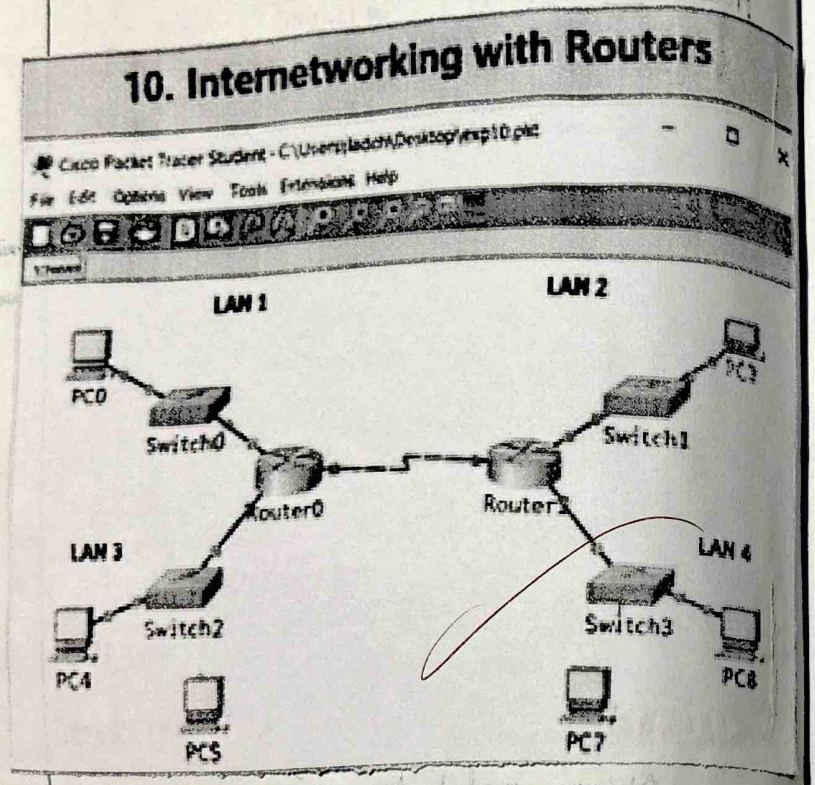
To implement classless IP subnetting using cisco packet tracer and establish communication between multiple subnets through routers.

Procedure:

1. Open cisco packet tracer.
2. Create a new network topology
3. Add routers, switches, and PCs.
4. Connect the devices using appropriate cables.
5. Divide the network 192.168.1.0/24 into smaller subn
6. Use /27 subnet masks as specified in the manu
7. Configure the router interface with suitable IP addresses.
8. Configure the switch ports.
9. Assign IP addresses and subnet masks to the PCs.
10. Configure the switch ports.
11. Enable router interfaces using no shutdown.
12. Test connectivity using ping
13. Observe packet transmission between subnets.

Results:

Thus, classless IP subnetting was successfully implemented in cisco packet tracer and communicat between the subnet was verified.



5/10

Ex No: 10

INTERNETWORKING WITH ROUTERS

Aim:

To design and configure an internetwork using routers in cisco packet tracer and verify communication between different networks.

Procedure:

1. Open cisco packet tracer.
2. Place one router and two PCs.
3. Connect the PCs to the router using copper straight-through cable.
4. Configure Router
5. Enter privileged mode:
   enable.
6. Enter configuration mode:
   config t
7. Configure Fast Ethernet 0/0
   interface FastEthernet 0/0
   ip address 192.168.10.1 255.255.255.0
   no shutdown
8. Configure Fast Ethernet 0/1:
   interface FastEthernet 0/1
   ip address 192.168.20.1 255.255.255.0
   no shutdown.
9. Configure PC0:

IP address : 192.168.10.2

Subnet Mask : 255.255.255.0

Gateway : 192.168.10.1

10. Configure PC1 :

IP address : 192.168.20.2

Subnet Mask : 255.255.255.0

Gateway : 192.168.20.1

11. Send a PDU from PC0 to PC1

12. Observe the packet transmission and acknowledgement.

Result :

Thus, an internetwork using a ~~network~~ router was successfully designed and configured and communication between networks was verified.

Sample Output:

Router # show ip route

R 192.168.20.0/24 [120/1]
via 192.168.10.2

Ping successful
Packet : sent = 4
Received = 4
Lost = 0

11. Routing at Network Layer (Static & RIP)

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#configure terminal
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#router rip
Router(config-router)#network 192.168.1.0
Router(config-router)#network 192.168.2.0
Router#show ip route
Codes: C - connected, S - static, R - RIP

C    192.168.1.0/24 is directly connected, GigE0/0
C    10.0.0.0/30 is directly connected, GigE0/1
S    192.168.2.0/24 [1/0] via 10.0.0.2
R    192.168.3.0/24 [120/1] via 10.0.0.2
Router#
```

Ex No: 11

Static Routing And RIP

Aim:

To simulate static routing and RIP (Routing Information Protocol) using cisco packet Tracer.

Procedure:

Part A - Static Routing

1. Open cisco packet tracer
2. Place two routers and suitable PCs
3. Connect the routers and PCs
4. Assign IP addresses to all interfaces
5. Configure router interfaces using ip address and no shutdown.
6. Add static router using.
   ip route <destination_network> <subnet-mask> <next-hop>
7. Verify the routing table using:
   show ip route.
8. Test connectivity using ping.

Part B - RIP

9. Configure the router interfaces
10. Enter RIP configuration mode:
    router rip
    version 2.

11. Add the connected networks
network <network_address>

12. Repeat the configuration on the other router.

13. Verify the routing table.

14. Ping a host in the remote network.

Result:

Thus, the static routing and RIP was successfully configured and the connectivity between different networks was verified.

Sample output:

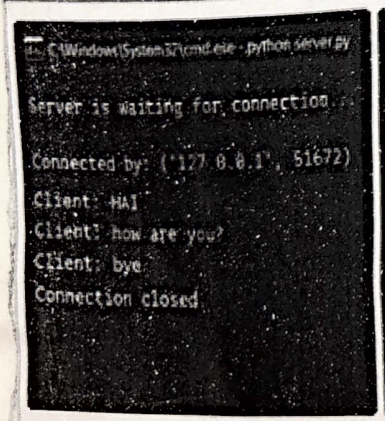
Server

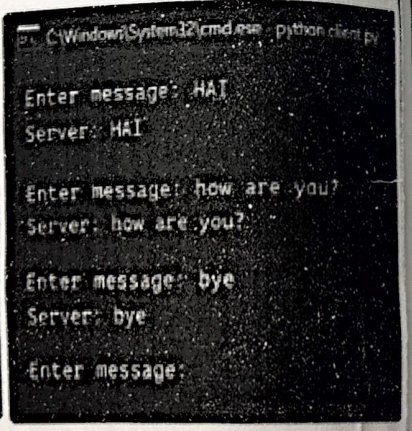
Server started...
Waiting for client...
Client connected!
Received: Hello Server
Echo: Hello Server.

Client

Connected to server
Enter message: Hello Server
Server replied.
Hello server.

12. End–End Communication (TCP/UDP Sockets)





CNP

Ex No: 12

TCP/UDP SOCKET COMMUNICATION

Aim:

To implement end-to-end communication at the transport layer using TCP/UDP sockets and develop a simple application.

Procedure:

Echo Client-server

1. Create a server program
2. Create a client program
3. Create a socket at the server
4. Bind the socket to an IP address and port.
5. Make the server listen for client connections
6. Accept the client connection.
7. The client sends a message
8. The server receives and echoes the message
9. Display the response at the client.

Chat program:

10. Create client and server socket programs
11. Establish a connection.
12. Read messages from the user.
13. Send the messages through the socket.
14. Receive and display messages at other end.
15. Continue communication until the user terminates the connection.

Result:

Thus, end-to-end communication was successfully implemented using socket-programming, and echo chat communication between client and server verified.

Sample Output:

Enter destination : 8.8.8.8

PING 8.8.8.8

Reply from 8.8.8.8

Sequence = 1 Time = 24 ms

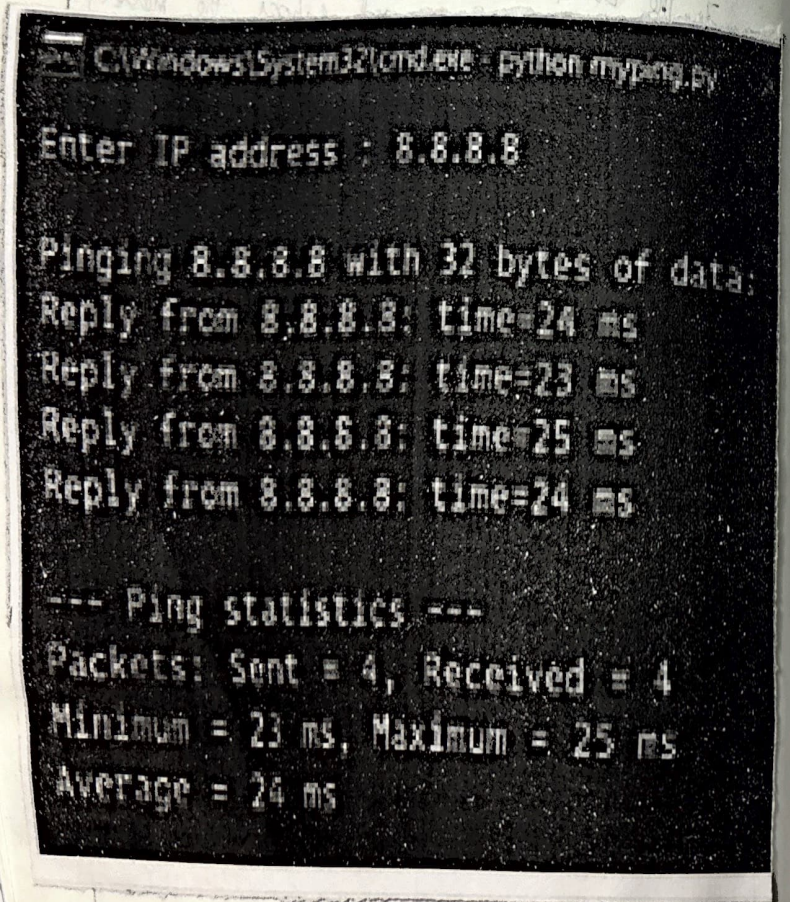
Reply from 8.8.8.8

Sequence = 3 Time = 25 ms

Packets sent : 3

Packets received : 3

Packets loss : 0%



13/10

Ex No : 13

OWN PING PROGRAM

Aim:

To implement a simple ping program using socket programming and verify the reachability of a destination host.

Procedure:

1. Create a program to accept a destination IP address / hostname.
2. Create a socket for network communication.
3. Generate a ICMP Echo Request.
4. Send the request to the destination.
5. Wait for an ICMP Echo Reply
6. Calculate the response time.
7. Display whether the host is reachable.
8. Repeat the process for multiple requests.
9. Display the process transmission statistics.

Result:

Thus, a simple ping program was successfully implemented and the reachability of the destination host was verified.

Sample Output:

RAW SOCKET PACKET SNIFFER

Packet Captured

Source IP : 192.168.1.10

Destination IP : 192.168.1.1

Protocol : TCP

Packet Length : 60 bytes.

14. RAW Socket Packet Sniffing

```
C:\Windows\System32\cmd.exe  python sniffer.py

Packet No: 1
Source IP       : 192.168.1.10
Destination IP  : 8.8.8.8
Protocol        : TCP
Source Port     : 54321
Destination Port : 80

--------------------------------

Packet No: 2
Source IP       : 8.8.8.8
Destination IP  : 192.168.1.10
Protocol        : TCP
Source Port     : 80
Destination Port : 54321
```

Ex No: 14

RAW SOCKET PACKET SNIFFING

Aim:

To implement packet sniffing using RAW sockets and analyze the packets recieved through the network interface.

Procedure:

1. Create a RAW socket using the socket programming interface.
2. Select the required network interface.
3. Enable packets reception through the RAW socket.
4. Capture incoming network packets
5. Read the packet header.
6. Extract important fields such as:
   - Source IP
   - Destination IP
   - Protocol
   - Packet length
7. Identify the protocol of the captured packet.
8. Display the packet information.
9. Continue capturing packets until the program is terminated.

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Result:

Thus, RAW sockets were successfully used to capture and analyse network packets.